Networking - Core Protocols & Communication

⚡ TCP vs UDP

📌 What are TCP and UDP?

- Both are **Transport Layer protocols** (Layer 4 in OSI, Transport Layer in TCP/IP).
- They define **how data is delivered** between two devices.
- Both use **ports** to identify services.

🔹 TCP (Transmission Control Protocol)

Characteristics

- **Connection-oriented** → establishes a connection before data transfer.
- **Reliable** → guarantees delivery (retransmits lost packets).
- **Ordered** → ensures packets arrive in the same sequence they were sent.
- **Error Checking** → uses acknowledgments (ACKs) and checksums.
- **Flow Control** → prevents sender from overwhelming receiver.

Data Unit

- Called a **Segment**.

Common Uses

- Web browsing (HTTP/HTTPS).
- Email (SMTP, IMAP, POP3).
- File transfers (FTP).

🔹 UDP (User Datagram Protocol)

Characteristics

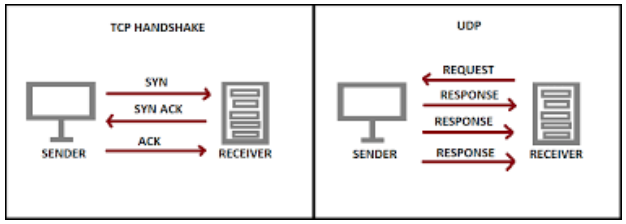
- **Connectionless** → no prior setup; just sends data.
- **Unreliable** → no guarantee of delivery.
- **No ordering** → packets may arrive out of order.
- **Lightweight** → minimal overhead, faster than TCP.
- **No flow control** → sends data regardless of receiver's state.

Data Unit

- Called a **Datagram**.

Common Uses

- Video/audio streaming.
- Online gaming.
- Voice over IP (VoIP).
- DNS queries.



🤝 TCP 3-Way Handshake

📌 What is the 3-Way Handshake?

- A **process used by TCP** to establish a reliable connection between a client and a server.
- Ensures that **both devices are ready** for communication and can synchronize sequence numbers.
- Called “3-way” because it involves **three messages**: SYN, SYN-ACK, ACK.

🔹 Steps of the 3-Way Handshake

Step 1: SYN (Synchronize)

- Client → Server
- The client sends a **SYN packet** (synchronize) to request a connection.
- This packet contains an **initial sequence number (ISN)** chosen by the client.

Message: “I want to connect, and my sequence number starts at X.”

Step 2: SYN-ACK (Synchronize + Acknowledge)

- Server → Client
- The server responds with **SYN-ACK packet**.
- This includes:
  - Its own sequence number (Y).
  - An acknowledgment number (X+1) confirming the client’s request.

Message: “I acknowledge your request (X+1), and here’s my sequence number Y.”

Step 3: ACK (Acknowledge)

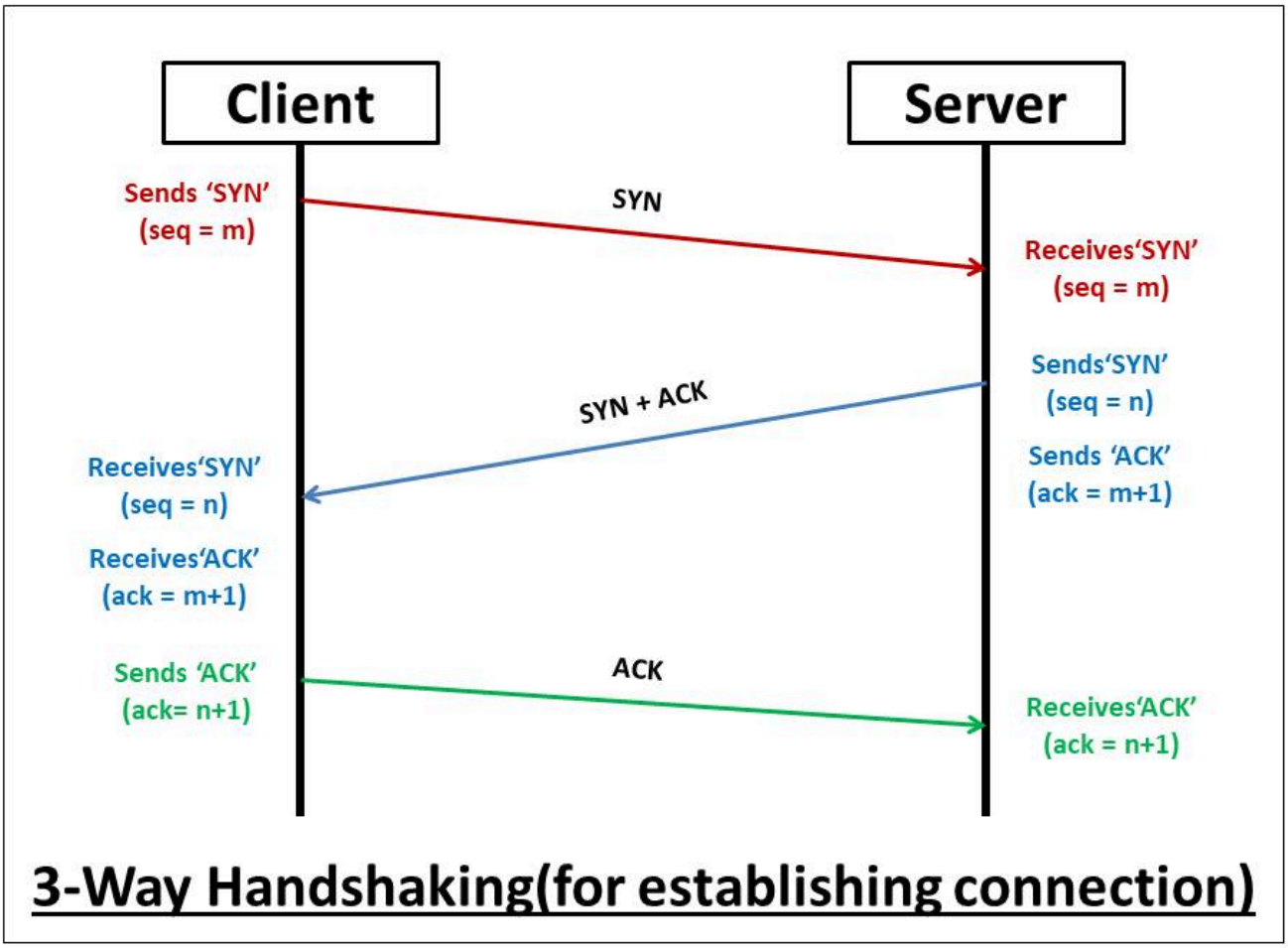
- Client → Server

- The client replies with an **ACK packet**.
- This acknowledges the server's sequence number (Y+1).

Message: "I acknowledge your sequence number (Y+1). Connection established."

📌 After Handshake

- Now the connection is established.
- Data transfer (segments) begins between client and server.



📡 ICMP (Internet Control Message Protocol)

📌 What is ICMP?

- ICMP is a **network layer protocol** (works with IP).
- Used for **sending control messages and error reporting**.

- Does **not carry user data** (like TCP/UDP), only network diagnostic information.

🔹 Common Uses of ICMP

1. Ping

- Checks whether a device (host) is reachable on the network.
- Works by sending an **ICMP Echo Request** and expecting an **ICMP Echo Reply**.

Example Flow:

1. Computer A sends Echo Request → Computer B.
2. Computer B replies with Echo Reply → Computer A.
3. Round-trip time (RTT) is measured.

Command Example:

```
ping 8.8.8.8
```

---

2. Traceroute

- Identifies the **path packets take** to reach a destination.
- Uses ICMP **Time Exceeded** messages (or sometimes UDP).

How it Works:

1. Sends packets with increasing **TTL (Time To Live)** values.
2. Each router on the path decreases TTL by 1.
3. When TTL reaches 0, the router replies with **ICMP Time Exceeded**.
4. This reveals the router's IP.
5. Process continues until the final destination is reached.

Command Example:

```
traceroute www.google.com   # Linux/macOS
tracert www.google.com      # Windows
```

🔹 ICMP Message Types (Common)

Type	Message	Use
0	Echo Reply	Response to ping
3	Destination Unreachable	Host, port, or network unreachable
5	Redirect	Informs better routing path
8	Echo Request	Ping request

Type	Message	Use
11	Time Exceeded	TTL expired (used in traceroute)

🌍 DNS (Domain Name System)

📌 What is DNS?

- DNS is like the **phonebook of the Internet**.
- Converts **human-readable domain names** (e.g., `www.google.com`) into **IP addresses** (e.g., `142.250.190.14`).
- Necessary because computers use IP addresses, while humans prefer names.

🔹 How DNS Resolution Works

When you type a domain into your browser:

1. **Browser Cache Check**
   - First, the browser checks if it already knows the IP from previous lookups.
2. **OS Cache Check**
   - If not in browser cache, the operating system checks its DNS cache.
3. **DNS Resolver (Recursive Resolver)**
   - If not cached, query goes to a DNS resolver (usually provided by ISP or public resolver like Google's `8.8.8.8`).
4. **Root DNS Server**
   - Resolver asks a **Root Server** (top-level) where to find info for `.com`, `.org`, `.net`, etc.
5. **TLD Server (Top-Level Domain)**
   - Root server points to the **TLD server** for `.com`.
   - TLD server responds with the address of the domain's **Authoritative Name Server**.
6. **Authoritative Name Server**
   - Stores the actual DNS record for the domain.
   - Example: For `www.google.com`, it returns the corresponding IP address.
7. **Response Sent Back**
   - Resolver sends the IP back to the browser.
   - Browser connects to that IP and loads the website.

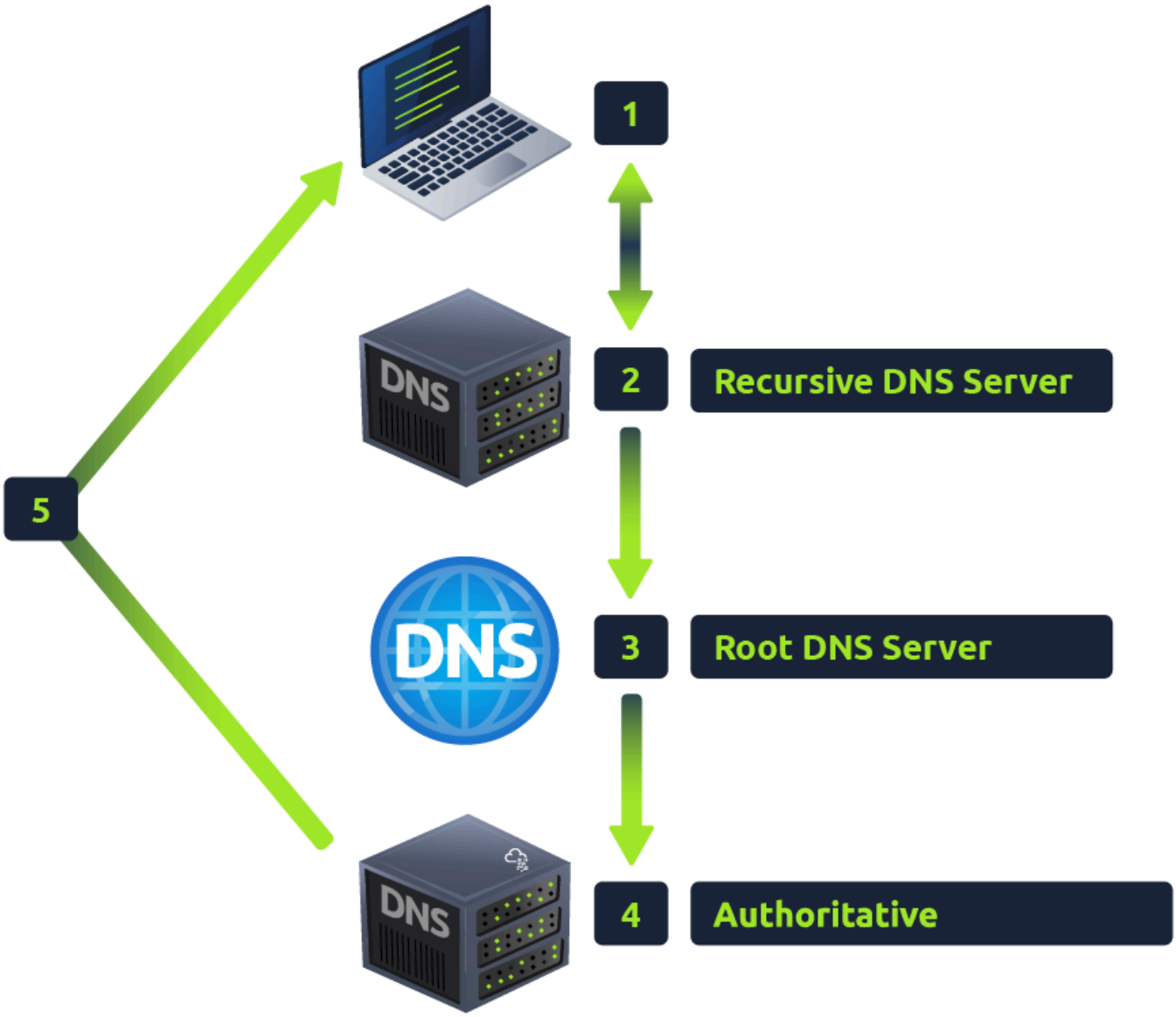
🔹 Types of DNS Records

Record	Purpose	Example
A	Maps domain to IPv4 address	`google.com → 142.250.190.14`
AAAA	Maps domain to IPv6 address	`google.com → 2a00:1450::142`
CNAME	Canonical name (alias)	`mail.example.com → example.com`

Record	Purpose	Example
MX	Mail exchange (email servers)	`gmail.com → mail.google.com`
TXT	Holds text info	SPF, DKIM records
NS	Nameserver for domain	`ns1.example.com`

🔹 DNS Hierarchy

- **Root Servers** (e.g., `.`)
- **TLD Servers** (`.com`, `.org`, `.net`)
- **Authoritative Servers** (store final record for domain)



DHCP (Dynamic Host Configuration Protocol)

📌 What is DHCP?

- **DHCP** is a protocol that **automatically assigns IP addresses** and network configuration to devices in a network.
- Saves administrators from manually assigning IPs to each device.
- Works at the **Application Layer** of TCP/IP (but manages Network Layer info).

🔹 Why DHCP is Needed

Without DHCP:

- Each device would need a **manually assigned IP address**.
- High chance of mistakes (duplicate IPs, wrong subnet mask).
  With DHCP:
- Devices receive correct configuration **automatically**.

🔹 What DHCP Assigns

1. **IP Address** – unique for each device.
2. **Subnet Mask** – defines network vs host portion.
3. **Default Gateway** – router IP for Internet access.
4. **DNS Servers** – for name resolution.
5. **Lease Time** – how long the device can use the IP before renewal.

🔹 DHCP Process (DORA)

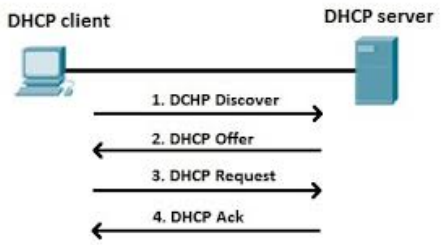
The process has **4 main steps**:

1. **Discover**
   - Client broadcasts a message:
     - “I need an IP address!”
2. **Offer**
   - DHCP server replies with an available IP + configuration details.
3. **Request**
   - Client sends a message back:
     - “I’d like to use this IP, please.”
4. **Acknowledge**
   - DHCP server confirms:
     - “Okay, this IP is yours for the lease time.”

👉 This is often remembered as **DORA (Discover, Offer, Request, Acknowledge)**.

🔹 Types of DHCP Allocation

1. **Dynamic Allocation**
   - IP assigned temporarily (with lease time).
   - Most common in home/office networks.
2. **Automatic Allocation**
   - DHCP server permanently assigns an IP to a device.
3. **Static/Manual Allocation**
   - Administrator binds a specific IP to a device's **MAC address**.
   - Common for servers, printers, routers.



HTTP & HTTPS Basics

📌 HTTP (Hypertext Transfer Protocol)

- Protocol used for **transferring web pages** on the Internet.
- Works on **Port 80** (default).
- Text-based and **not secure** → data (including passwords) sent in plain text.

📌 HTTPS (HTTP Secure)

- Secure version of HTTP.
- Works on **Port 443**.
- Uses **SSL/TLS encryption** → ensures data is encrypted, authenticated, and safe from eavesdropping.

📬 FTP, SMTP, POP3, IMAP

🔹 1. FTP (File Transfer Protocol)

📌 What it is

- Protocol for **transferring files** between client and server.
- Works at the **Application Layer**.

📌 Ports

- **Port 21** → Command/control connection.
- **Port 20** → Data transfer (in Active Mode).

📌 Modes of FTP

1. **Active Mode:** Server initiates the data connection back to the client.
2. **Passive Mode:** Client initiates both control and data connections (commonly used, works better with firewalls).

📌 Uses

- Uploading/downloading files.
- Managing files on web servers.

🔹 2. SMTP (Simple Mail Transfer Protocol)

📌 What it is

- Protocol for **sending emails** from a client to a mail server or between mail servers.

📌 Ports

- **Port 25** → Default, used for server-to-server transfer.
- **Port 587** → Submission from clients (with authentication).
- **Port 465** → Secured SMTP with SSL (legacy but still in use).

📌 Uses

- Outgoing mail delivery.
- Relaying mail between servers.

🔹 3. POP3 (Post Office Protocol v3)

📌 What it is

- Protocol for **retrieving emails** from a mail server.
- Downloads emails to the client and **removes them from the server** (by default).

📌 Port

- **Port 110** (unencrypted).
- **Port 995** (POP3S, encrypted with SSL/TLS).

📌 Uses

- Good for single-device email access.

- Emails are stored locally, freeing server storage.

🔹 4. IMAP (Internet Message Access Protocol)

📌 What it is

- Protocol for **retrieving and managing emails**.
- Unlike POP3, emails remain on the server by default.

📌 Port

- **Port 143** (unencrypted).
- **Port 993** (IMAPS, encrypted with SSL/TLS).

📌 Uses

- Allows synchronization across multiple devices.
- Supports folders, message states (read/unread).

📊 SNMP (Simple Network Management Protocol)

📌 What is SNMP?

- A protocol used for **monitoring and managing network devices**.
- Works at the **Application Layer**.
- Allows administrators to gather information, configure, and sometimes control network equipment remotely.

🔹 Components of SNMP

1. **SNMP Manager**
   - Central system (usually a monitoring server).
   - Sends queries and receives responses.
   - Example: Network monitoring tools (Nagios, Zabbix, SolarWinds).
2. **SNMP Agent**
   - Runs on the network device (router, switch, printer, server).
   - Collects device info (CPU usage, traffic, uptime).
   - Responds to manager's requests.
3. **MIB (Management Information Base)**
   - A structured database of information that devices can report.
   - Defines what metrics (objects) are available (e.g., interface status, error rates).

🔹 How SNMP Works

- The SNMP manager sends **requests** (GET, SET) to the agent.
- The agent replies with the requested information.
- Devices can also send **Traps** (alerts) to the manager when certain events occur (e.g., interface down).

🔹 SNMP Operations

Operation	Description
GET	Manager asks agent for info (e.g., “What’s your uptime?”).
SET	Manager changes a configuration on the agent (e.g., reset interface).
GETNEXT	Retrieves the next object in the MIB tree.
TRAP	Agent sends an alert to the manager (e.g., device failure).

🔹 SNMP Versions

- **SNMPv1** → First version, very basic, community string authentication (not secure).
- **SNMPv2c** → Added improvements (bulk transfers, error handling).
- **SNMPv3** → Added **encryption & authentication**, making it secure.

🔐 SSH & Telnet

🔹 1. Telnet (Telecommunication Network)

📌 What it is

- One of the earliest protocols for **remote login** into another computer over a network.
- Works at the **Application Layer**.

📌 Characteristics

- Uses **Port 23** (default).
- Sends all data (including usernames & passwords) in **plain text** → not secure.
- Provides a **command-line interface** for remote control.

📌 Use Cases (Historical)

- Remote management of servers and network devices.
- Debugging and testing network services.
- Rarely used today due to lack of encryption.

🔹 2. SSH (Secure Shell)

📌 What it is

- A **secure replacement for Telnet**.
- Provides **encrypted communication** over the network.
- Also works at the **Application Layer**.

📌 Characteristics

- Uses **Port 22** (default).
- Encrypts all communication using **cryptographic algorithms** (public/private keys, symmetric encryption).
- Provides **secure remote login** and secure file transfer (SCP, SFTP).

📌 Use Cases

- Secure remote server administration.
- Secure tunneling.
- Secure file transfers.

SSL/TLS Encryption

Purpose:

- Secure communication over the network.
- Provides **confidentiality, integrity, and authentication**.
- Commonly used in HTTPS, FTPS, SMTPS, IMAPS, POP3S.

---

1. SSL vs TLS

Feature	SSL	TLS
Full Name	Secure Sockets Layer	Transport Layer Security
Versions	SSL 2.0, SSL 3.0	TLS 1.0, 1.1, 1.2, 1.3
Security	Outdated, vulnerable	Modern, more secure
Handshake	Less secure handshake	Stronger handshake & crypto

> Note: TLS is the successor of SSL. Today, SSL is mostly obsolete.

---

2. Key Components

1. **Encryption (Confidentiality):**
   - Symmetric encryption (AES, ChaCha20) used for data transfer.
   - Public key (asymmetric) used for secure key exchange.
2. **Authentication:**
   - Digital certificates (X.509) verify server identity.
   - Certificate Authority (CA) signs the certificate.

3. **Integrity:**
   - Message Authentication Codes (MACs) ensure data is not tampered.

---

3. TLS Handshake Process

1. **Client Hello:**
   - Client sends supported TLS version, cipher suites, and a random number.
2. **Server Hello:**
   - Server picks TLS version & cipher, sends its certificate (and public key).
3. **Key Exchange:**
   - Client and server agree on a session key (Diffie-Hellman / RSA).
4. **Session Key Established:**
   - Symmetric encryption begins for all further communication.
5. **Finished Messages:**
   - Both verify handshake integrity, secure channel is established.